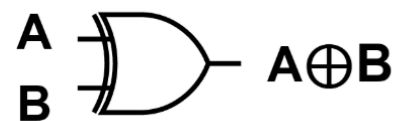


Experiment 3: Implementation of Advanced gates (X-OR and X-NOR) with their corresponding equation.

Theory: In digital electronics, X-OR (Exclusive OR) and X-NOR (Exclusive NOR) are known as advanced logic gates. These gates are widely used in arithmetic circuits, error detection, comparators, and digital communication systems.

X-OR (Exclusive OR) Gate: An X-OR gate produces a HIGH (1) output only when the inputs are different. If both inputs are the same, the output remains LOW (0). If one input is HIGH and the other is LOW, the output is HIGH. If both inputs are HIGH or both are LOW, the output is LOW.

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

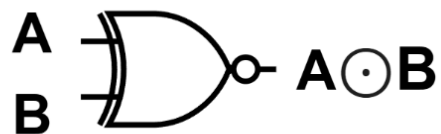


1 = on
0 = off

Boolean Equation: $Y = A \oplus B = A'B + AB'$. This equation shows that the output is true when either A is false and B is true, or A is true and B is false.

X-NOR (Exclusive NOR) Gate: An X-NOR gate is the complement of the X-OR gate. It produces a HIGH (1) output when both inputs are the same. If both inputs are LOW or both are HIGH, the output is HIGH. If the inputs are different, the output is LOW.

A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1



1 = on
0 = off

Boolean Equation: $Y = A \odot B = AB + A'B'$. This equation shows that the output is true when both inputs are equal.

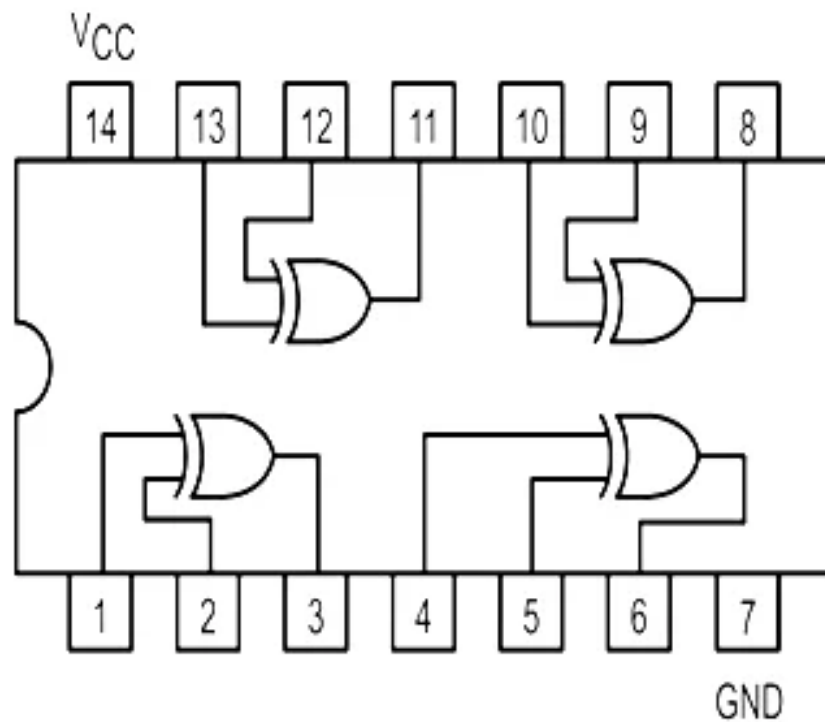
Apparatus Required:

1. Breadboard
2. Connecting Wires

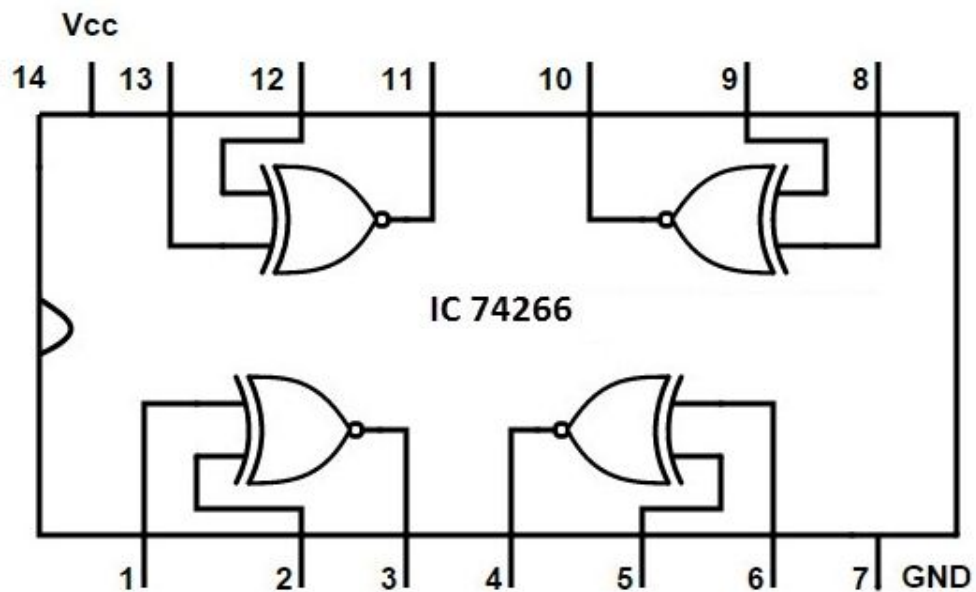
3. IC 7486 (X-OR Gate)
4. IC 74266 (X-NOR Gate)
5. IC 7404 (NOT Gate)
6. IC 7408 (AND Gate)
7. IC 7432 (OR Gate)
8. Power Supply (+5V)
9. LEDs

IC Structure:

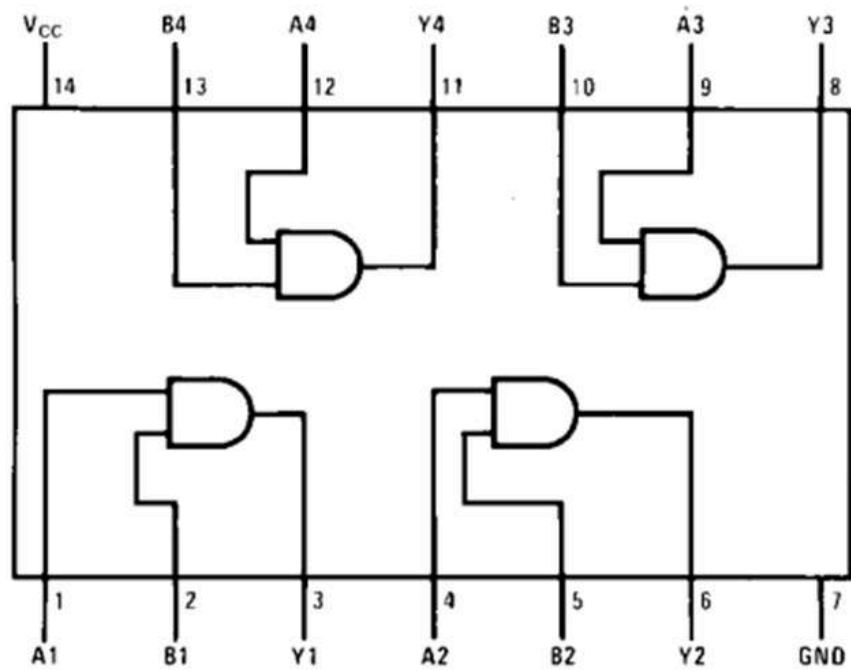
1. IC 7486 (X-OR Gate):



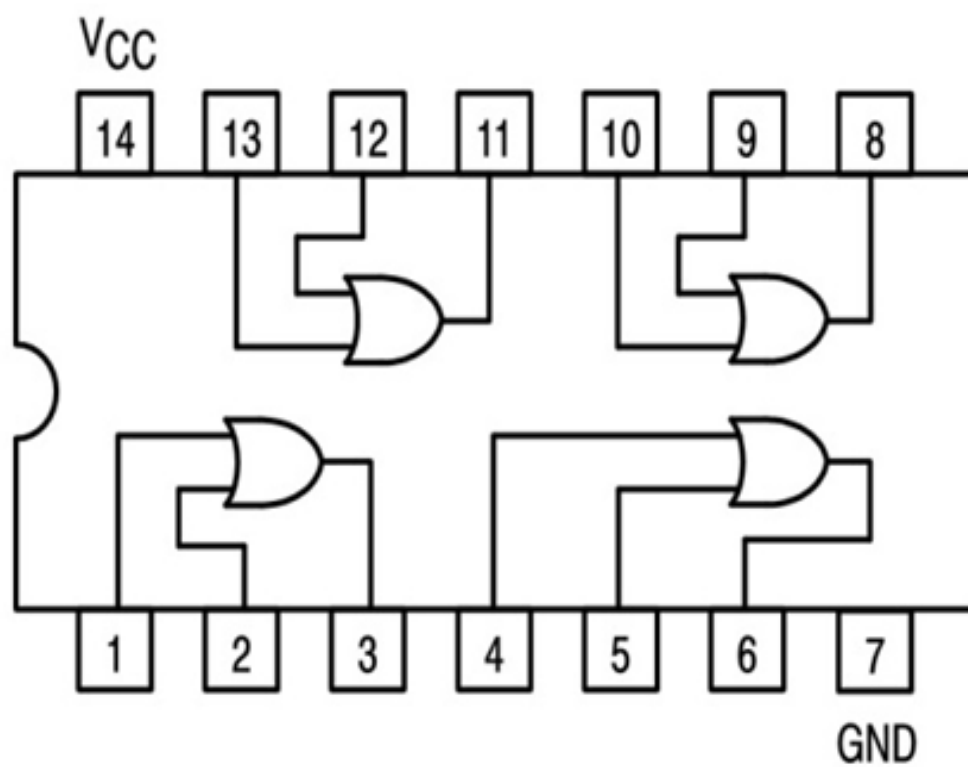
2. IC 74266 (X-NOR Gate):



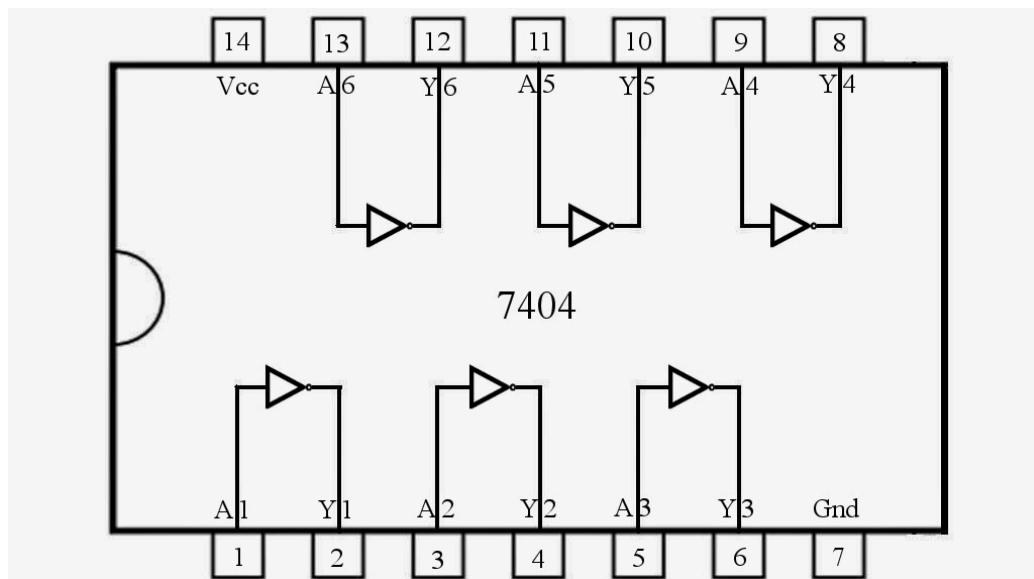
3. IC 7408 (AND Gate):



4. IC 7432 (OR Gate):



5. IC 7404 (NOT Gate):



Conclusion: Thus, the X-OR and X-NOR gates are successfully studied and implemented. Their logical behavior is verified using truth tables and Boolean expressions. These gates play a crucial role in digital systems such as adders, subtractors, and comparators.